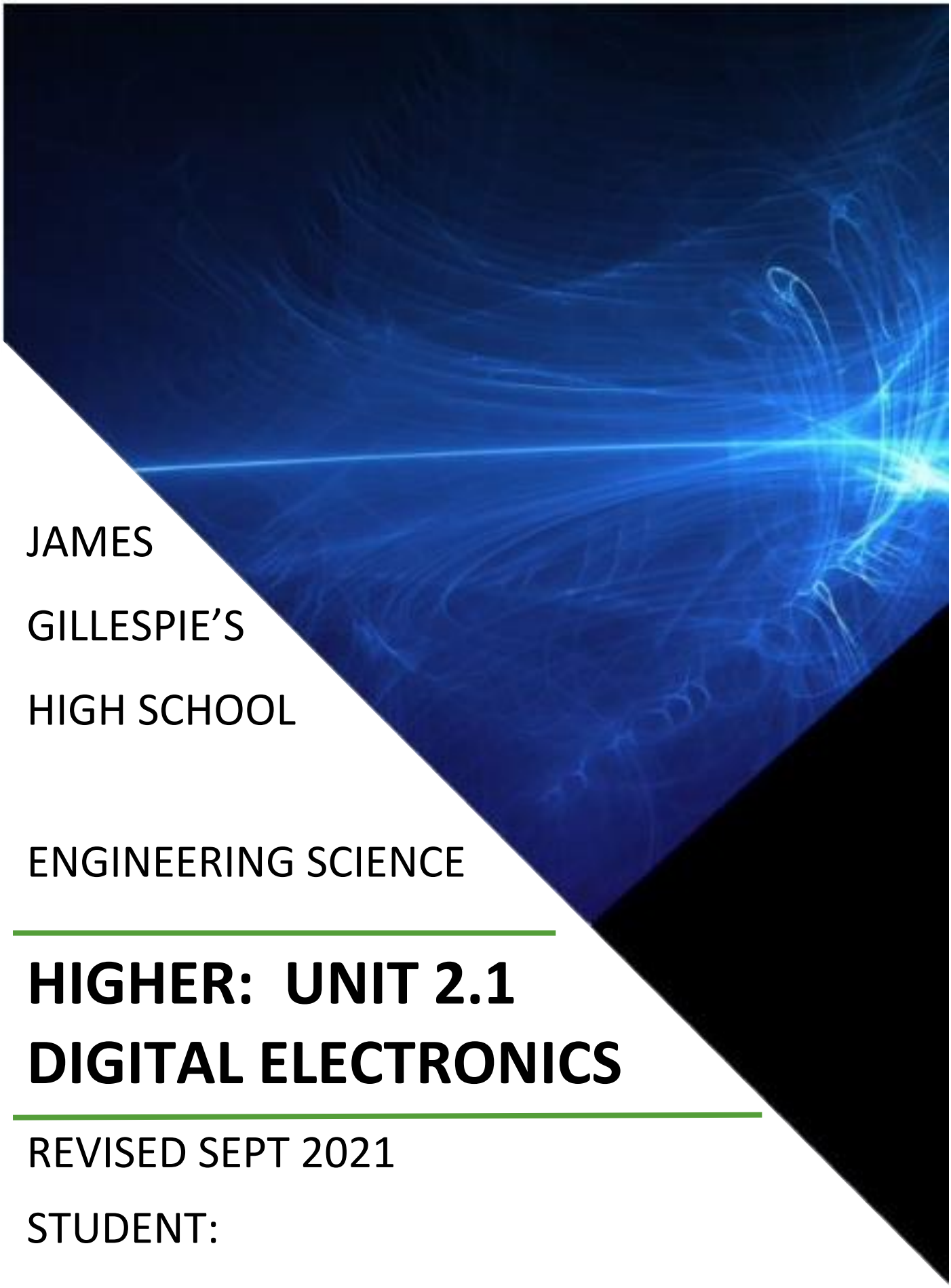


JAMES

GILLESPIE'S

HIGH SCHOOL

ENGINEERING SCIENCE

HIGHER: UNIT 2.1 DIGITAL ELECTRONICS

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STUDENT:

YEAR: TEACHER:

Contents

Learning intentions 2

For digital electronic control: 2

Success criteria 2

Skills, knowledge and understanding for course assessment: 2

Basic logic gates 3

Truth tables 4

Structure of truth tables 4

Truth tables for individual logic gates 5

Circuit simulation software 5

Combinational logic systems 7

Analysing a combinational logic system 7

NAND gates within Integrated Circuits (IC) 9

Drawing NAND equivalent circuits 10

Boolean algebra 12

Basic logic gates and their Boolean representations 12

NAND gate 12

Deriving the Boolean expression for a circuit 14

Deriving the Boolean expression from a truth table 18

Designing a circuit from its Boolean equation 19

Designing circuits 21

Worked example: 22

Integrated circuits 25

Advantages of Integrated Circuits: 25

CMOS vs TTL 25

TTL 25

CMOS 25

Identifying ICs 26

IC circuit diagrams 28

Exam style practice questions 31

Learning intentions

For digital electronic control:

- To implement the logic functions AND, OR, NOT, NAND, NOR, EOR and combinations with up to four inputs
- To convert logic systems to NAND equivalents
- To develop Boolean expressions from truth tables, logic diagrams or circuit specifications
- To construct truth tables and logic diagrams from written specifications

Success criteria

Skills, knowledge and understanding for course assessment:

In order to track your progress of learning, consider the above learning intentions and record your progress. This will also help with revision as you will be able to see what areas you need to work on. Your teacher will review this record page after speaking to you about your self-evaluation.

☐ I can implement the logic functions.

☐ I can convert logic systems to NAND equivalents.

☐ I can develop Boolean expressions.

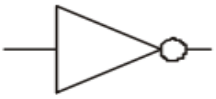
☐ I can construct truth tables and logic diagrams from written specifications.

Basic logic gates

There are six logic gates; these are the NOT, AND, OR, NAND, NOR and XOR. You met the first three of these during National 5. In this course we deal with the all six, as shown below.

1 In your jotter, copy all the logic gates with an explanation of what each does.

NOT



When a high (1) signal inputs, a low signal outputs (0).

AND



When both signals are high (1), output is high (1). Otherwise, output is low (0).

NAND



When both signals are high (1), output is invested to low (0) low. The reverse happens only when both input signals are low to get a high (1) output.

OR



When one or both of the inputs is high (1), the output is high (1).

NOR



The opposite of an OR gate: Only when none of the inputs are high (1), the output is high. Only operates when both input rails are low (0).

XOR



'X' stands for exclusive. When only one of the inputs is high, the output is high. Output rail low (0) when both input rails are high (1). When there are two input rails, it only operates with one and never both.

Truth tables

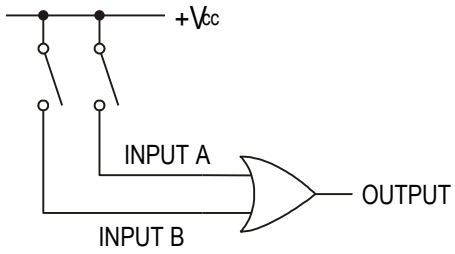
The easiest way to represent how each gate behaves is to make use of truth tables. A truth table shows all possible combinations of inputs to and outputs from a logic gate.



Digital signals can be at a HIGH voltage level or a LOW voltage level. In logic circuits a LOW signal is said to be at logic '0' a HIGH signal at logic '1', known as digital logic

Structure of truth tables

Consider the inputs to a logic gate as two switches.



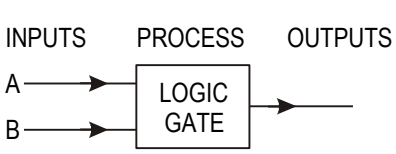
It is possible for each switch to be in one of two positions, either on (1 High) or off (0 Off). These positions are known as input states. The connection lines can be referred to a 'rails'. When creating a truth table, it may be easier to assign a lettered code to each input rail (A, B, C) and any process points as mid alphabet (M, N) with the output signal generally referred to as Z.

For two inputs, there are only 4 possible combinations of input states: A-(low) & B-(low)off; A-(low) & B-(high); A-(high) & B-(low); A-(high) & B-(high). For three inputs, there are 8 possible combinations.

In general, for **'n'** inputs there are **2^n** possible combinations of input states.

Input signals come from a variety of sources - a switch from a keyboard; a bar code reader; a temperature sensor; another part of a computer.

Output signals can have a variety of destinations - a monitor; a modem; an alarm; another part of a computer.



Input A	Input B	Output
0	0	0
0	1	1
1	0	1
1	1	1

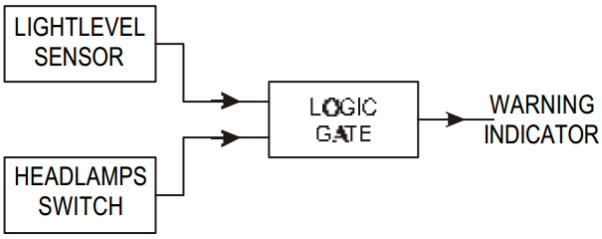
2

a) Draw a truth table for all 6 logic gates shown on the previous page.

b) How many combinations of input states would there be for a 6 input system?

c) Write down the 8 possible combinations of input states for a 3 input system.

d) To avoid accidents at times of poor visibility, a warning indicator in a car operates if the light level is too low (logic level 0) and the headlamps are switched off. Which logic gate would be needed for this operation?

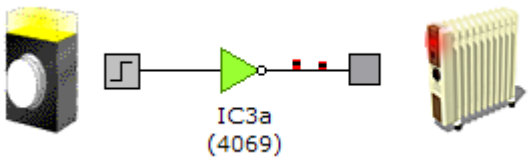


Truth tables for individual logic gates

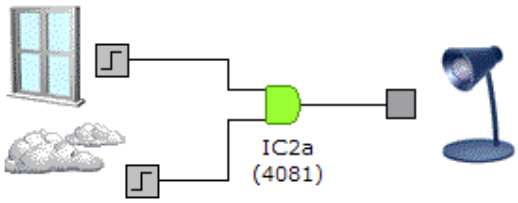
Circuit simulation software

Logic circuits can be investigated and simulated using circuit simulation software such as Yenka. Animated inputs and outputs allow for more realistic simulation of engineered products. Try creating each of these circuits and check it behaves as you expect.

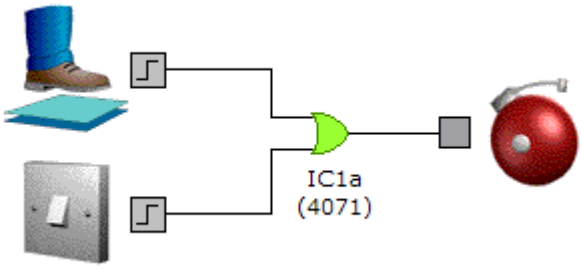
1. NOT gate



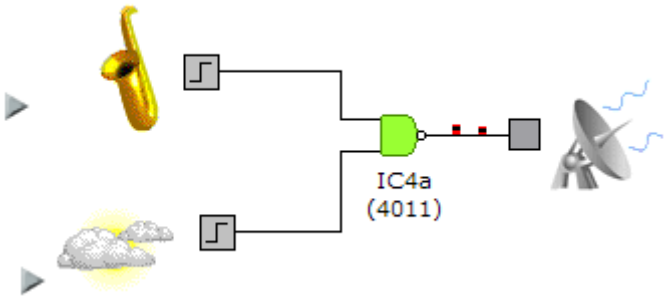
2. AND gate



3. OR gate



4. NAND gate



3

a) Why is the NOT gate sometimes called an 'INVERTER'

b) When is the output of an OR gate high?

c) When is the output of an AND gate high?

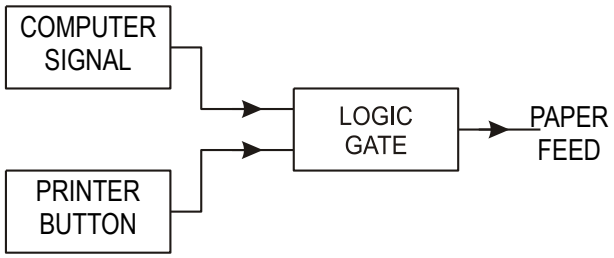
d) When is the output from a NAND gate high?

e) The truth table below shows the output conditions for the various combinations of input conditions for AND, NAND and OR gates. Complete the table and discuss how the output of the NAND gate compares with the output of the AND and OR gates.

Inputs		AND	NAND	OR
0	0			
0	1			
1	0			
1	1			

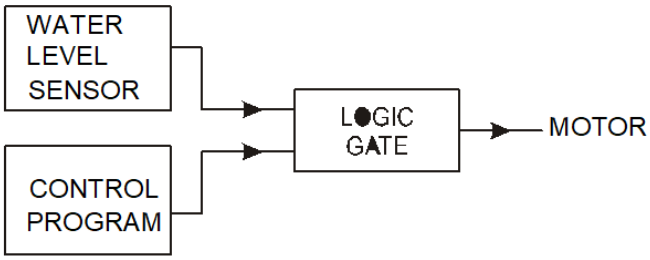
4

The paper can be fed through a computer printer either by pressing the button on the printer (line feed) or by sending a signal from the computer. State the name and sketch the logic gate which should be should be used for this operation.



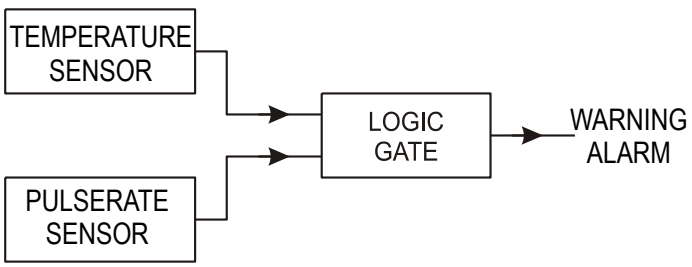
5

The motor in a washing machine should not operate until a high signal is sent from the control program and the water level in the drum is high enough. State the name and sketch the logic gate which should be should be used for this operation.



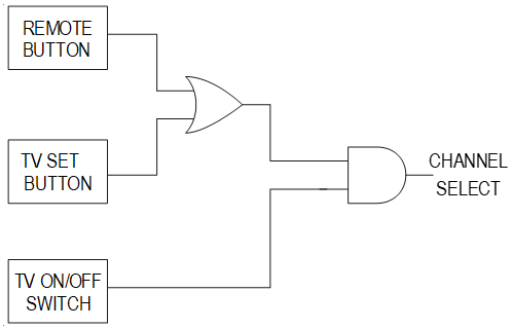
6

In the maternity unit of a hospital, the temperature and pulse rate of premature babies has to be continually monitored. A warning alarm should sound if either the temperature or the pulse rate of the baby falls too LOW. State the name and sketch the logic gate which should be should be used for this operation.



Combinational logic systems

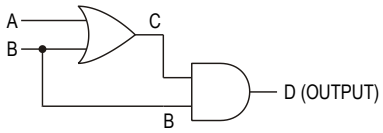
It is often necessary to use more than one gate to perform a decision process. For example, on some TV sets it is possible to change the channel either by pressing the channel select button on the TV or by pressing the channel select button on the remote control. The channel will only change however, if the TV set is switched on. i.e. The channel will change if the TV set button is pressed **OR** the remote button is pressed **AND** the TV set is on. A logic diagram for this is shown below.



Analysing a combinational logic system

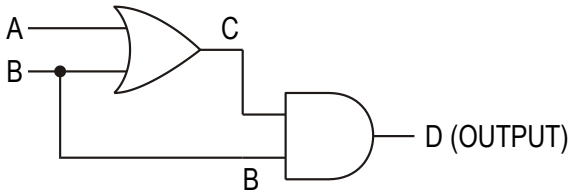
The output of these circuits is determined by the combination of input signals which is present at that time (i.e. the output state does not depend on the sequence of the input signals). If we need to know how such a circuit works, we can analyse it according to the possible combinations of inputs and create a truth table.

If this logic diagram is given:



Step 1

Label all the different points on the circuit, including inputs and outputs. Notice that one input serves two gates, this is quite common in logic circuits and both gates should be labelled accordingly.



Step 2

Draw a truth table for the circuit, with a different column for each letter. Remember to determine the number of input rows needed, consider the number of input conditions. In this case we have two input, therefore 4 lines on the truth table ($2^2 = 4$).

Step 3

Determine what goes into column C. To do this, consider the OR gate on its own and enter findings into column C.

INPUT A	INPUT B	C	OUTPUT D
0	0	**0**	
0	1	**1**	
1	0	**1**	
1	1	**1**	

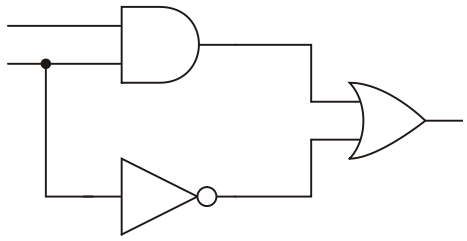
Step 4

Determine what goes into output column D. To do this consider the AND gate with inputs from C and B enter findings into column D. Your final answer should look like this. A more complicated system may need several stages before you finish, but if you work systematically through the circuit in the way shown, considering one gate at a time, you should arrive at the correct answer.

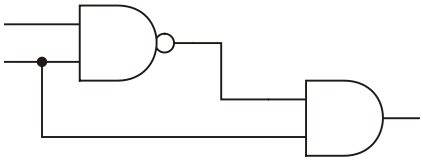
INPUT A	B	C	OUTPUT **D**
0	0	0	**0**
0	1	1	**1**
1	0	1	**0**
1	1	1	**1**

7 Copy each combination logic gate and complete a truth table shown below.

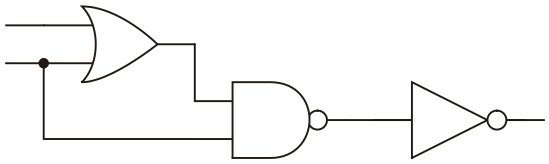
a.



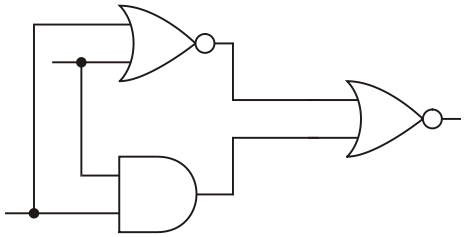
b.



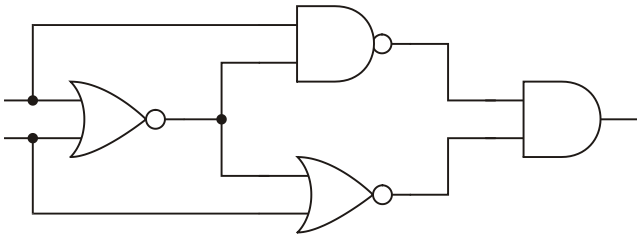
c.



d.



e.

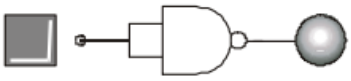


.

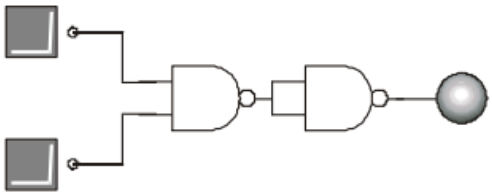
NAND gates within Integrated Circuits (IC)

NAND gates are called functionally complete, as it is possible to make other logic gates by combining NAND gates. Using these as universal gates can be very beneficial; it can provide cheaper solutions to a problem as only one chip set would need to be bought, compared to four or five different ICs of different gates. Using only ICs containing NANDs will also take up less space, saving vital space in an electronic system.

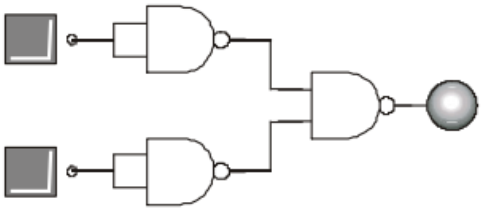
Use truth tables or circuit simulation software to work out which logic gate the following NAND configurations represent. **Copy** these into your jotter as you will need to remember these.



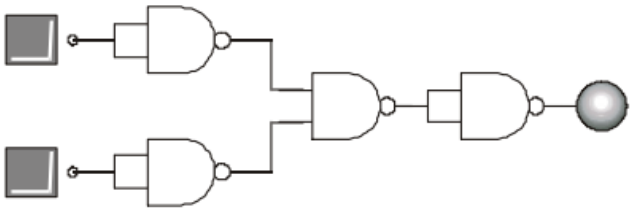
NOT gate



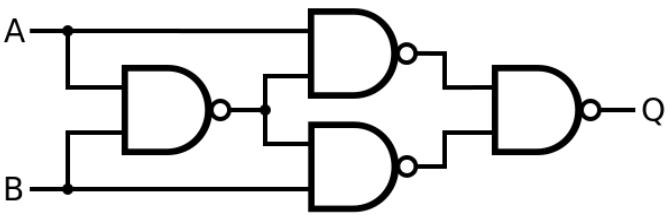
AND gate



OR gate



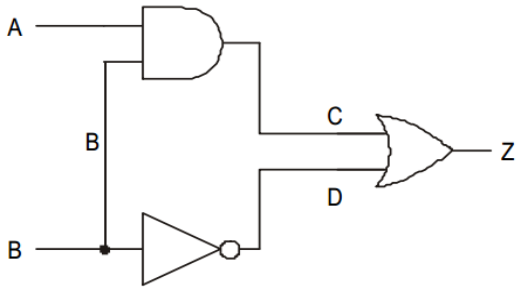
NOR gate



XOR gate

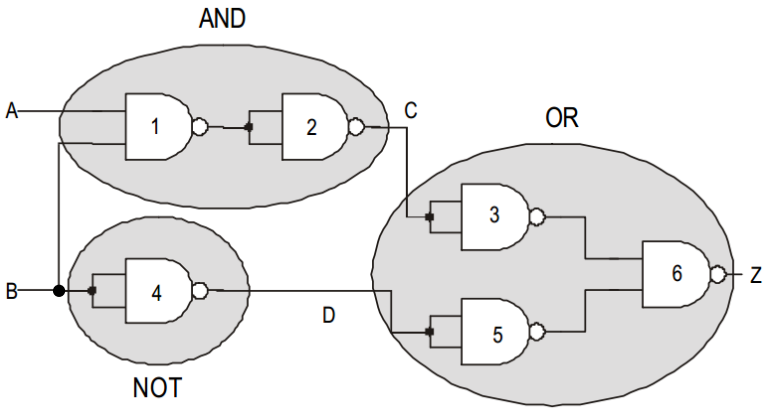
Drawing NAND equivalent circuits

When presented with a logic circuit, it is important to understand how to convert it into a NAND equivalent circuit. Consider the circuit shown:



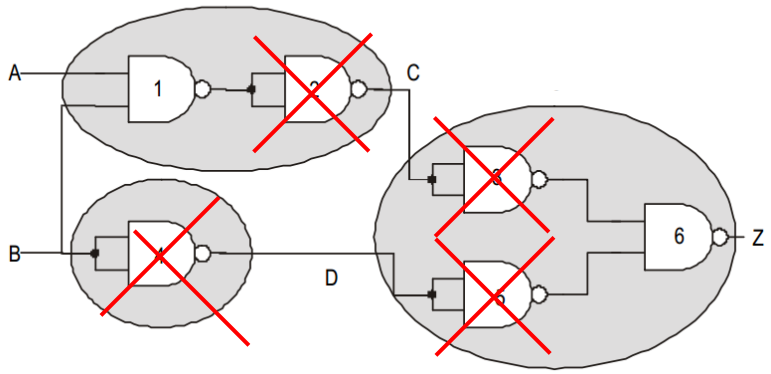
Step 1

Redraw the circuit using NAND gates.

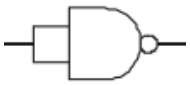


Step 2

If there are two NOT gates together then remove them. To save you redrawing the whole circuit, you can just cross them out.



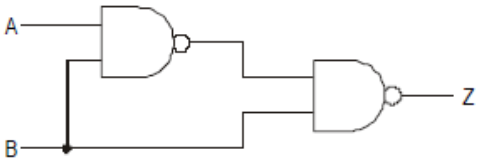
Remember: The NAND equivalent of a NOT gate is…



Note how one input is split in two, there are NOT two separate input signals.

Step 3

Redraw your final circuit.

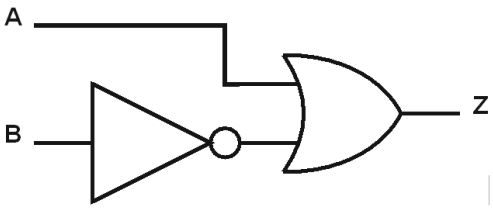


The final circuit has only two gates whereas the original circuit started with three gates, each of a different type. There are obvious implications in terms of cost for manufacturers if they can reduce logic circuits to ones with fewer gates as well as enabling them to use one type of gate. The original circuit would have required three IC's and this final circuit would only require one IC.

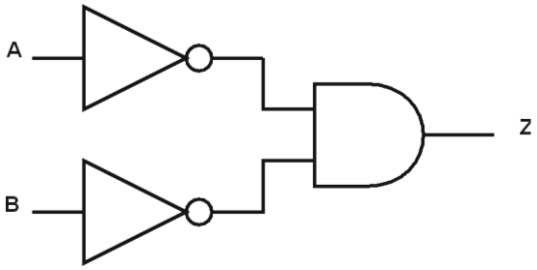
This method may not reduce the number of gates used on every occasion but it should reduce the number of IC's used.

8 The following logic diagrams are constructed from basic gates. Using the method shown, copy the original logic gate and construct an equivalent circuits using NAND gates only.

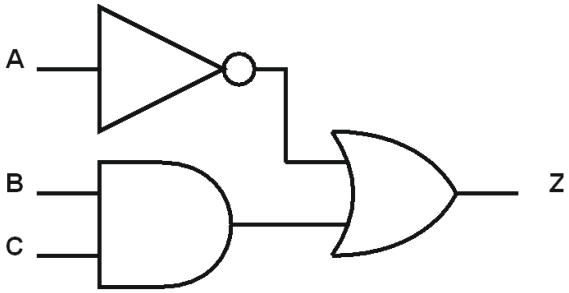
a)



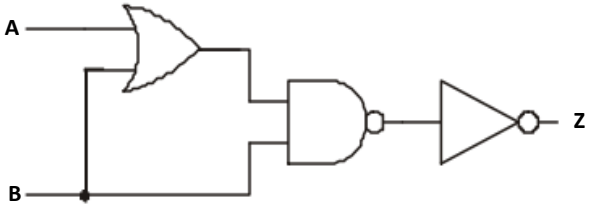
b)



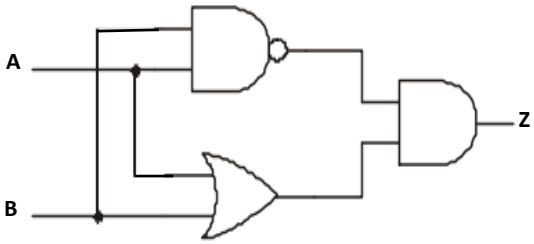
c)



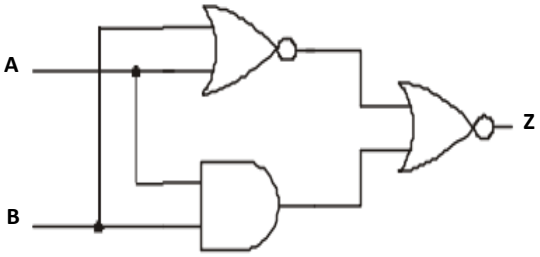
d)



e)



f)

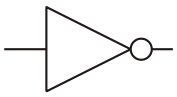
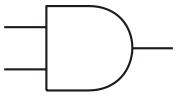
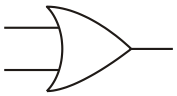


Boolean algebra

Boolean algebra is a form of algebra which is useful for analysing and designing logic systems. It was invented by George Boolean in 1854 and is still used today. Usually A, B, C… are used as inputs. Z, Y, X… are used to refer to outputs.

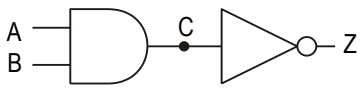
You met Boolean algebra in National 5 and this page will reintroduce you to the basics, before we move on to deriving Boolean expressions for circuits. It is essential you understand Boolean and don't confuse it with normal algebra symbols.

Basic logic gates and their Boolean representations

LOGIC SYMBOL	BOOLEAN EXPRESSION	DESCRIPTION	TRUTH TABLE
NOT 	$\overline{A} = Z$	NOT **A** EQUALS **Z**	A | Z 0 | 1 1 | 0
AND 	A.B = Z	**A** AND **B** EQUALS **Z**	A | B | Z 0 | 0 | 0 0 | 1 | 0 1 | 0 | 0 1 | 1 | 1
OR 	A+B = Z	**A** OR **B** EQUALS **Z**	A | B | Z 0 | 0 | 0 0 | 1 | 1 1 | 0 | 1 1 | 1 | 1

NAND gate

The NAND gate is made up from a combination of an AND gate followed by a NOT gate.



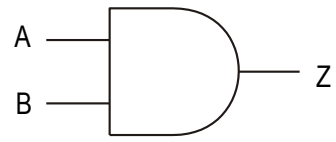
The signal at point C would be $A \bullet B$. This signal is then inverted by the NOT gate to give.

$$\overline{A \bullet B} = Z$$

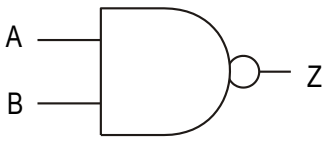
This reads as output Z is equal to A AND B all NOT.

9 Copy each logic gate and write down the Boolean expression:

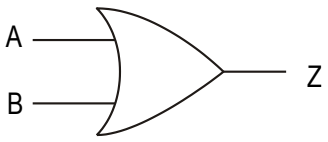
a)



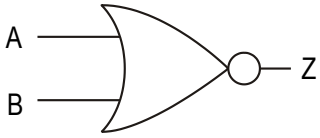
b)



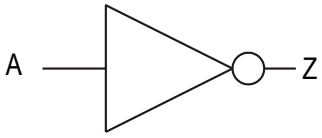
c)



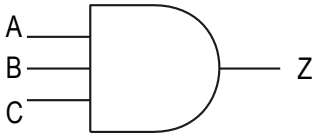
d)



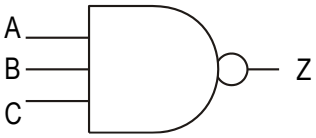
e)



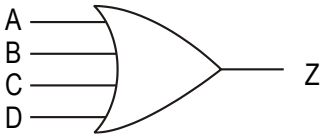
f)



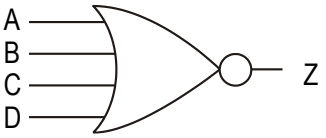
g)



h)

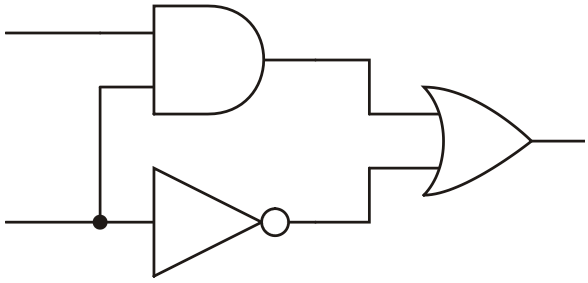


i)



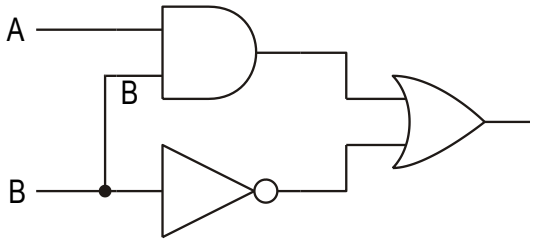
Deriving the Boolean expression for a circuit

Consider the following circuit:



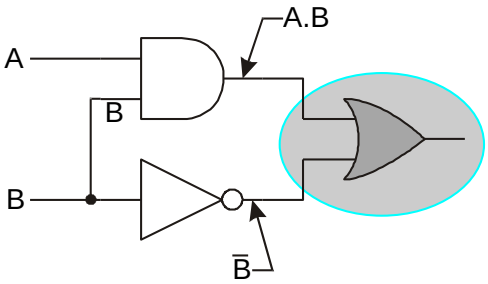
Step 1

Label the inputs on the left-hand side of the diagram (If one input serves more than one gate ensure that the input is labelled accordingly).



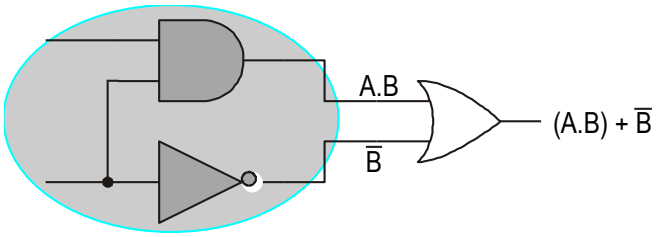
Step 2

Consider each gate in turn. Use the Boolean notation (operators) to give the output of each gate in terms of its input. Write on the appropriate expression after each gate.



Step 3

When outputs from other gates are inputs to a further gate, treat the expressions as you would any other equation and make use of brackets (if necessary). Then write on the appropriate expression after the next gate and so on until you reach the final output.



Step 4

Write down the final Boolean expression for the network.

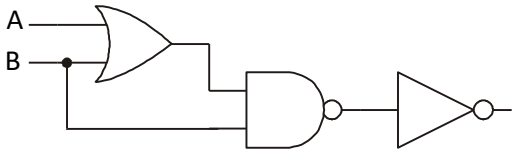
$$(A \bullet B) + \overline{B} = Z$$

(Since normal rules of algebra hold, $\overline{B} + (A \bullet B) = Z$ would also be correct)

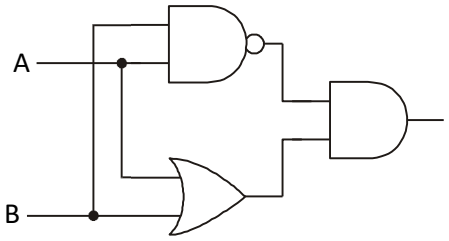
10

Copy each logic circuit and derive the Boolean expression:

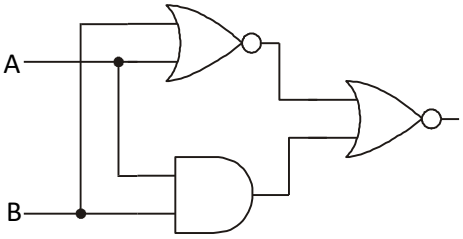
a)



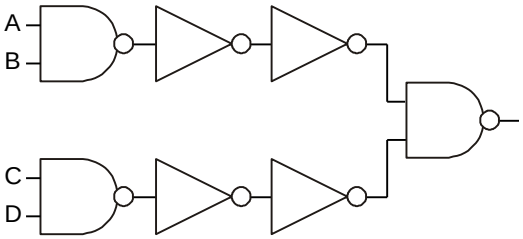
b)



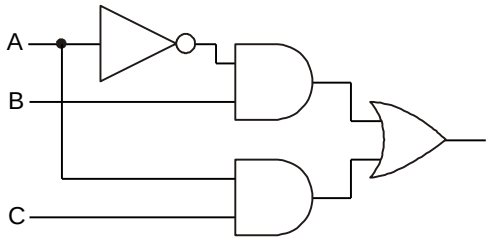
c)



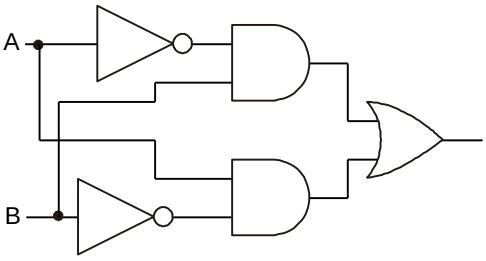
d)



e)



f)



11

Draw a gating arrangement to illustrate $Z = A \bullet B + C$

12

a) Draw a logic diagram to yield $D = B + C$

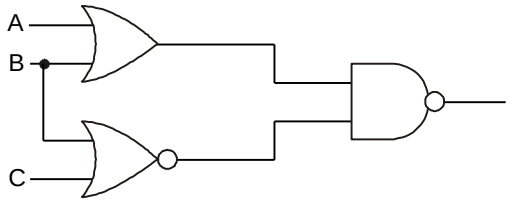
b) Develop it to obtain $Z = A \bullet D$

c) Write a Boolean equation for the overall behaviour.

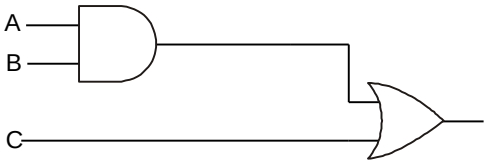
13

Derive the Boolean equation and the truth table the following arrangements:

a)



b)



14

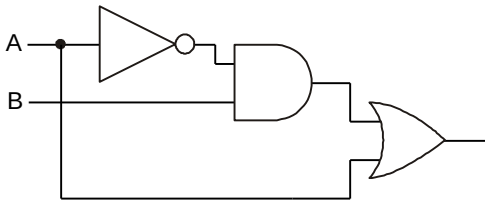
Copy each pair of circuits shown below:

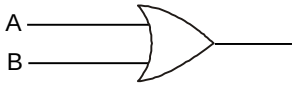
(i) Write a Boolean expression for each of these circuits;

(ii) By constructing a truth table for each of them, show that they are equivalent;

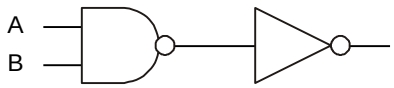
(iii) Draw the equivalent truth table arrangements using only 2-input NAND gates.

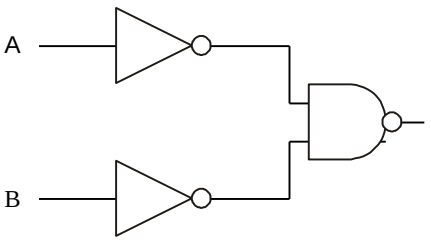
a)



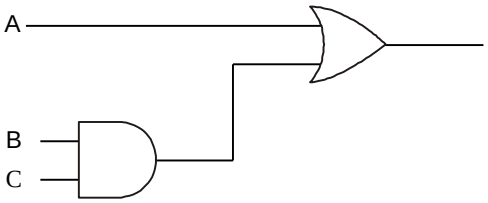


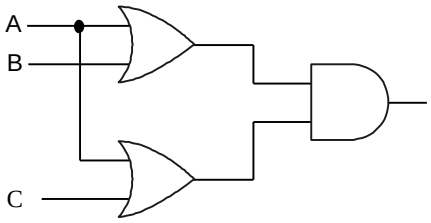
b)





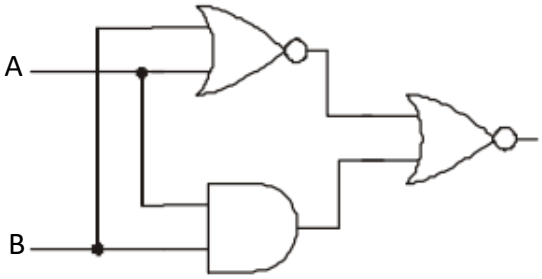
c)



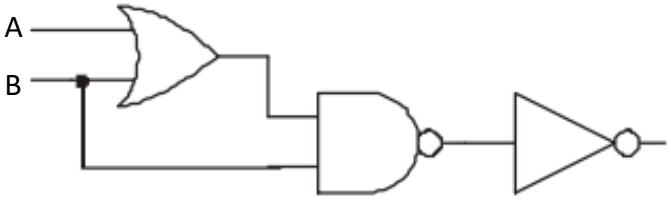


15 copy each circuit below, derive the Boolean expression and associated truth table. Construct the equivalent circuit using only NAND gates.

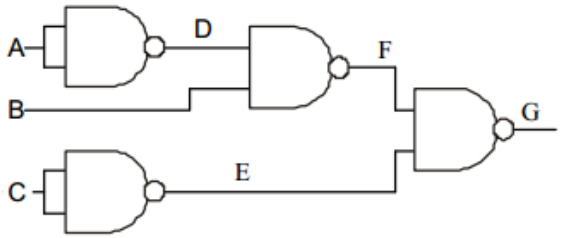
a)



b)



16 The diagram below shows part of an industrial control system having three inputs A, B and C with an output G.



(i) How many different input conditions are possible in this system?

(ii) What is the function of gates on the left?

(iii) Complete a truth table including columns which show the states of D, E, F and G.

(iv) Why is the design made up entirely of NAND gates?

(v) Give an alternative design using AND, OR and NOT gates.

Deriving the Boolean expression from a truth table

Consider the truth table given:

A	B	Z
0	0	1
0	1	0
1	0	1
1	1	1

Step 1

Note each combination that will give you a '1' at the output and write the Boolean expression for this line at the side of the Truth table, next to the line that it applies to.

A	B	Z	
0	0	1	$\overline{A} \bullet \overline{B}$
0	1	0	
1	0	1	$A \bullet \overline{B}$
1	1	1	$A \bullet B$

Step 2

Join the equations by putting an OR sign between each to give the final Boolean expression.

$$(\overline{A} \bullet \overline{B}) + (A \bullet \overline{B}) + A \bullet B = Z$$

17 Write the Boolean equation for the following truth tables.

a)

A	B	Z
0	0	0
0	1	1
1	0	1
1	1	0

b)

A	B	C	Z
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

c)

A	B	Z
0	0	1
0	1	0
1	0	1
1	1	0

d)

A	B	C	Z
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

Designing a circuit from its Boolean equation

Consider the following equation:

$$\overline{A} \bullet \overline{B} + A \bullet \overline{B} + A \bullet B = Z$$

Step 1

Draw inputs A and B (and any other input variables) on the left-hand side of the page.

A

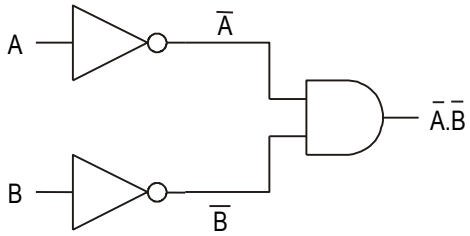
B

Step 2

Start by considering the first term:

$$\overline{A} \bullet \overline{B}$$

It can been seen from the equation that both inputs require to be inverted. This is achieved by passing signals from A and B through NOT gates. The output from these signals are then used as inputs to an AND gate (as signified by the dot between the A and B in the Boolean expression). Write this on the output line of the AND gate.

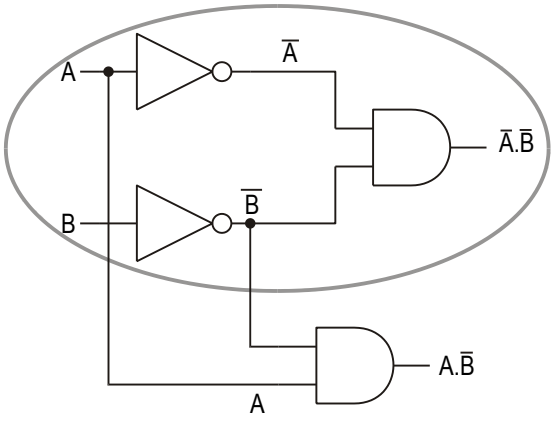


Step 3

Consider the next term:

$$A \bullet \overline{B}$$

It can be seen from this term that input A does not require to be inverted but input B does. Since we already have both input A and B bar available we can amend the circuit diagram accordingly by feeding these inputs to a second AND gate.

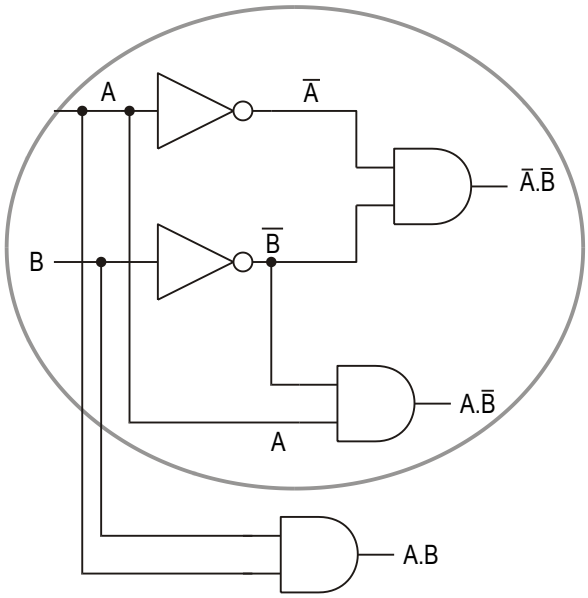


Step 4

Consider the final term:

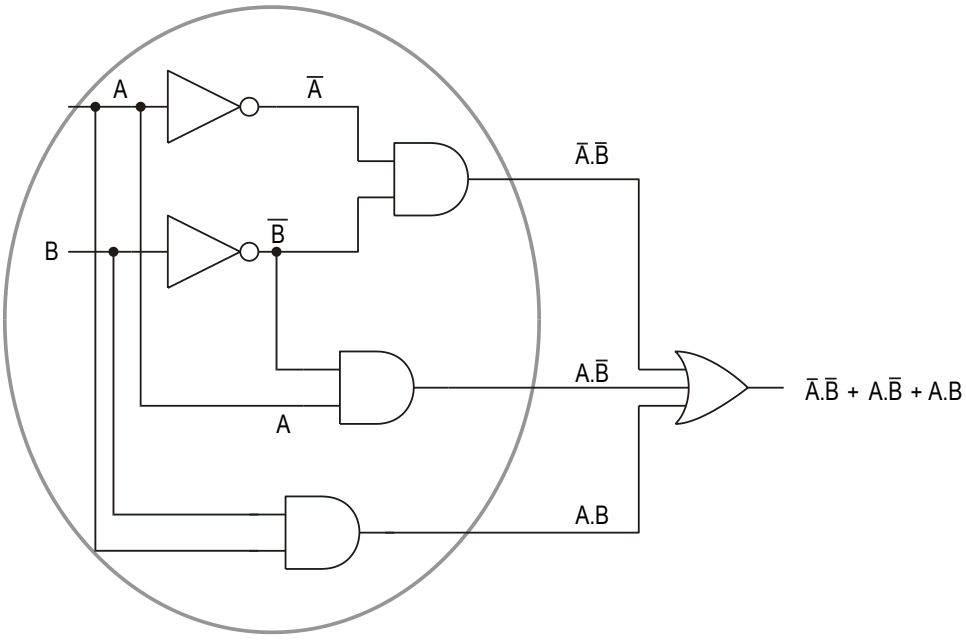
$$A \bullet B$$

Again both inputs are available to us. Neither of the inputs is inverted so the signals are taken from the original inputs and are fed to a third AND gate.



Step 5

Finally we use the outputs from the three AND gates as inputs to a three input OR gate to arrive at the final solution.



This is the completed circuit. Deriving circuits from Boolean expressions can lead to large and complex diagrams, make sure you follow these steps carefully to design the circuit correctly.

18 For the following Boolean equations, draw the correct logic circuit arrangement.

a) $A + (A \cdot B) = Z$

b) $A + \bar{B} + C = Z$

c) $(A \cdot B) + (A \cdot C) = Z$

d) $(A \cdot B) + (\bar{A} \cdot C) + (B \cdot C) = Z$

e) $(A \cdot B \cdot C) + (D \cdot E) + (F \cdot G) = Z$

19 Develop a Boolean equation and draw a logic circuit diagram containing AND, OR and NOT gates to yield the truth table shown.

A	B	C	Z
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

Designing circuits

As problems become more and more difficult it is not always possible to go from the question to the answer in one or two steps, when that is the case the following set of rules should be followed.

When designing a system to suit a need you should proceed in the following order.

1. Describe the problem clearly in words.
2. Write out a Truth Table for the system.
3. Derive the Boolean expression from the Truth Table.
4. Simplify this expression if possible.
5. Draw a logic circuit diagram for the system using AND, OR and NOT gates.
6. Convert the circuit to NAND gates only.

It is entirely possible that not every problem will require all of these steps to be followed, however this will be a useful guide for most.

Worked example:

K

This system has two inputs (4 possible combinations) and three outputs (green light, red light and bell).

Step 1

Describe the problem clearly in words.
Let the temperature sensor be input A and the fault acknowledge switch as input B.

Green light is on if: A is 0 AND B is 0

OR A is 0 AND B is 1

The second condition occurs if the engineer acknowledges a fault when there is no fault.

The bell rings when: A is 1 AND B is 0

The red light is on when: A is 1 AND B is 0

OR A is 1 AND B is 1

Step 2

Write out a truth table for the system.

Inputs A	B	green light	red light	alarm bell
0	0	1	0	0
0	1	1	0	0
1	0	0	1	1
1	1	0	1	0

Step 3

Derive the Boolean expression from the Truth Table.

Inputs A	B	green light		red light		alarm bell	
0	0	1	$\overline{A} \bullet \overline{B}$	0		0	
0	1	1	$\overline{A} \bullet B$	0		0	
1	0	0		1	$A \bullet \overline{B}$	1	$A \bullet \overline{B}$
1	1	0		1	$A \bullet B$	0	

Step 4

Simplify the expressions if possible.
green light - $(\overline{A} \bullet \overline{B}) + (\overline{A} \bullet B) = Z$

It can be seen that the output does not depend on the state of B since the output is HIGH when B is either HIGH or NOT HIGH i.e. the state of the green light depends only on the state of A, hence the expression can be simplified to

green light - $\overline{A} = Z$

red light - $(A \bullet \overline{B}) + (A \bullet B) = Z$

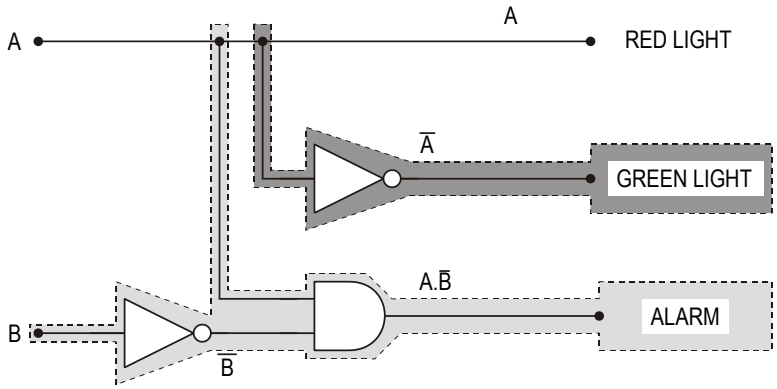
Similarly it can be seen that the red light is ON irrespective of the state of B - the output only depends on the state of A hence this can be simplified to

red light - $A = Z$

alarm - $A \bullet \overline{B} = Z$ (can't be simplified further)

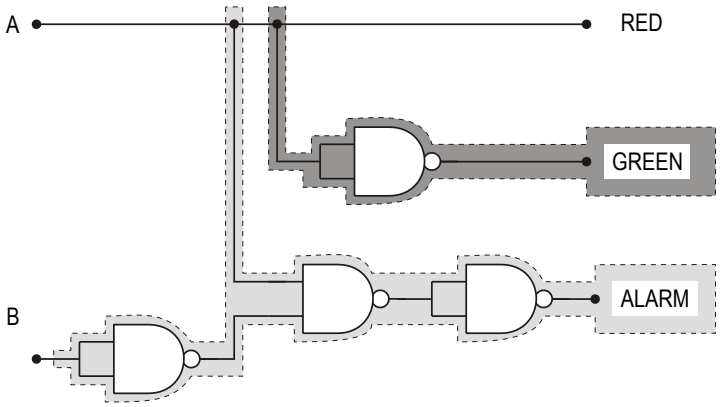
Step 5

Draw a logic circuit diagram for the system using AND, OR and NOT gates.
Start with the expression that has fewest terms/gates (in this case *A=Z,* no gates!)



Step 6

Convert the circuit to NAND gates only.
(Look for double inverted signals which can be cancelled out.)



20 A domestic burglar alarm system is designed such that a bell will operate when the power switch is closed and a pressure switch under a carpet is closed or a switch is opened as a window is lifted. Assuming all switches to be logic state zero (0) when open.

(i) Draw a logic diagram for the design, allocating capital letters to the inputs to each gate and to the output to the bell.
(ii) Prepare a truth table for the design. Your table must be headed by the appropriate letters
(iii) Show by use of a logic diagrams how you would modify the circuit to use only NAND gates.

21

An electric guillotine must be adequately guarded. In order to safeguard the operator the machine has two switches, A and B, set about one metre apart, both of which need to be pressed before the machine will operate.

Design a logic circuit that will give a green light if both switches are pressed. If only one switch is pressed a red light should come on. Assume that the switches return a 0 when not pressed and a 1 when pressed.

22

A given logic circuit has two logic inputs A and B. It is required to produce two logic outputs X and Y according to the following rules:

a) X is to be at logic 1 if (A OR B) but NOT (A AND B) are at logic 1.
b) Y is to be at logic 1 if (A AND B) but NOT (A OR B) are at logic 1.

Use the previous method to design an appropriate logic circuit.

23

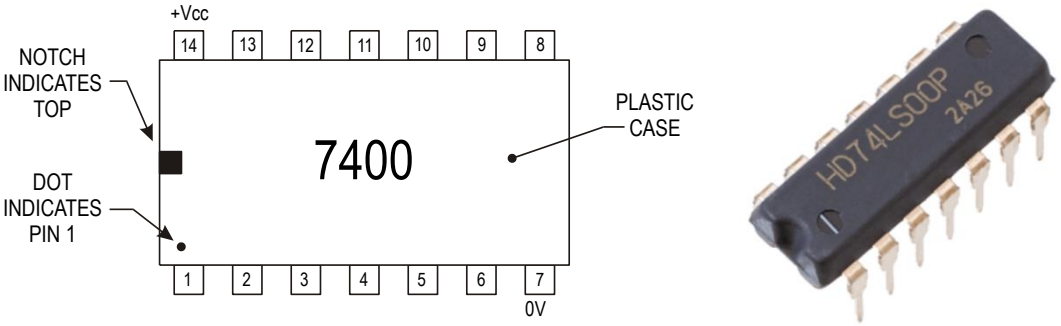
A garage door is operated by a motor which is controlled by three switches. The motor runs either when a pressure pad switch, A, in the drive is closed and a light dependent resistor switching circuit, B, is simultaneously activated by the car's headlights *or* when the key switch, C, in the garage door is operated.

a) Prepare a truth table for all possible combinations of switching conditions for switched A, B and C. Take switch open as logic 0.
b) From the truth table, prepare a logic diagram using the least number of gates.
c) Redesign your circuit to use only 2-input NAND gates.

Integrated circuits

In digital electronic circuits, logic gates are normally manufactured onto an integrated circuit (IC). An IC can, and more often does, contain more than one logic gate.

IC's can be found in a wide variety of packages. The most common is the plastic DIL (Dual In Line) pack, where the silicon chip is enclosed in the plastic case with connecting pins to provide connections to the appropriate gates. Here the pins run down either side of the plastic package as shown below. The number of pins can be 8, 14, 16, 18, 20, 24, 40 and even 60.



Pin numbering starts with pin 1 which is always below the notch or identifying circle. The numbering then goes advances in an anti-clockwise direction.

Advantages of Integrated Circuits:

Integrated circuits have the following advantages over discrete components.

1. They are smaller and lighter.
2. They are cheaper than discrete components due to manufacturing techniques and scale of integration (the number of components on each chip).
3. More reliable and easier to replace.

CMOS vs TTL

There are two main logic families of IC chips: CMOS and TTL. It is important to know the differences and when one is preferable to the other.

CMOS

Complementary Metal Oxide Silicon. These use MOSFETS to perform logic operations.

Key advantages:

- Operates in a wide range of DC voltages from 3V to 15V.
- Low power consumption.

Key disadvantages:

- Slow switching speed.
- Not static resistant.

TTL

Transistor-Transistor Logic. These use bipolar transistors to perform logic operations.

Key advantages:

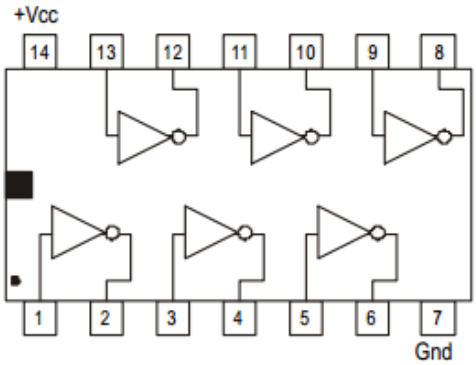
- Fast switching speed.
- Resistant to static.

Key disadvantages:

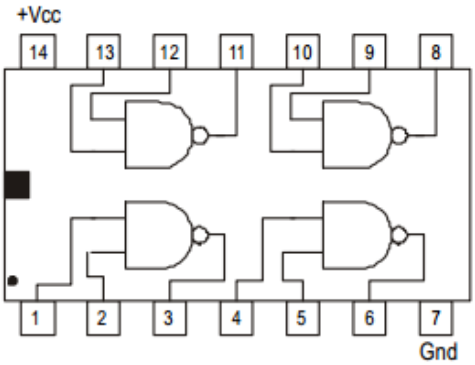
- Requires an operating voltage of 5V (or within 0.25V of 5V).
- Higher power consumption than CMOS devices.

Identifying ICs

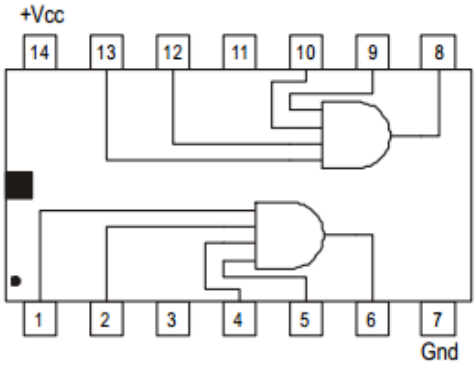
This page shows the pin layouts for ICs relevant to this course. You do not need to memorise them, however you should be confident in using them.



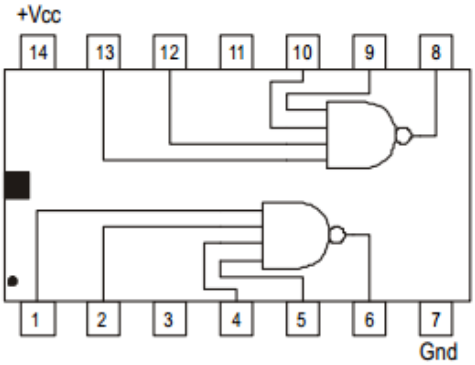
7404



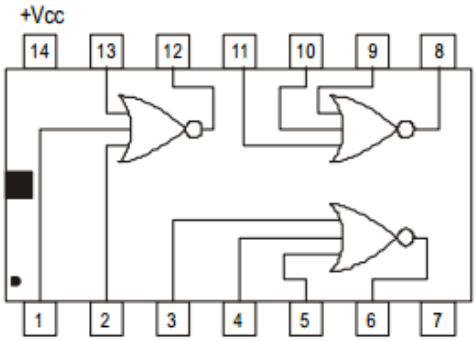
7400



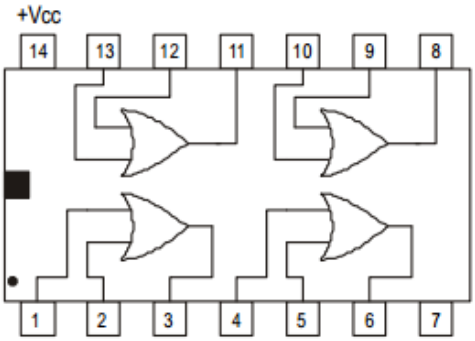
7421



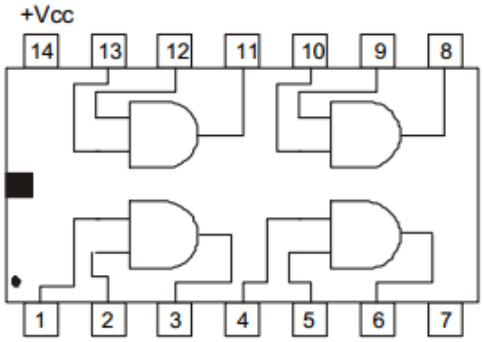
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7427



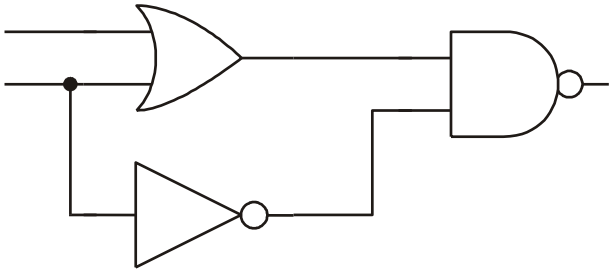
7432



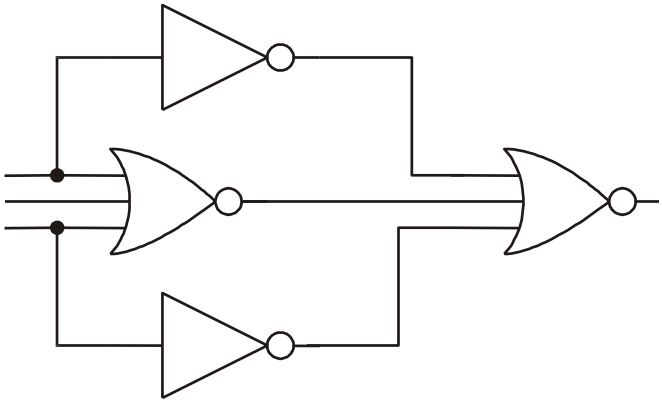
7408

24 Which IC's would be required to construct the following circuits?

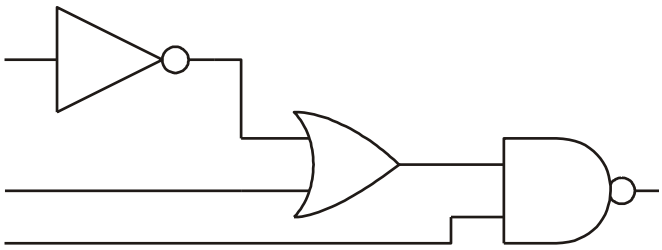
a)



b)

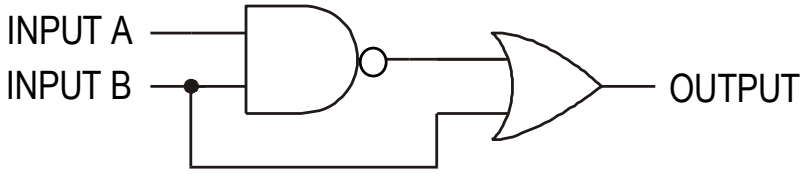


c)

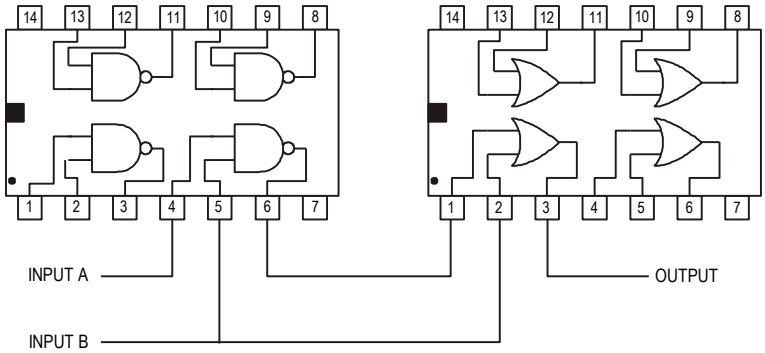


IC circuit diagrams

The following logic circuit could be constructed using IC's.

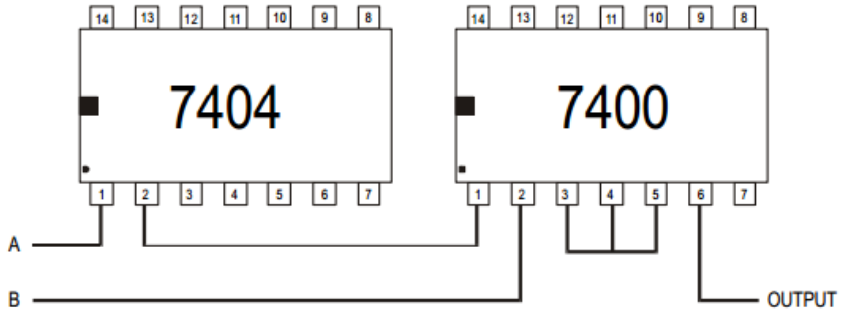


Since the gates within an IC are identical, any one of them can be used.

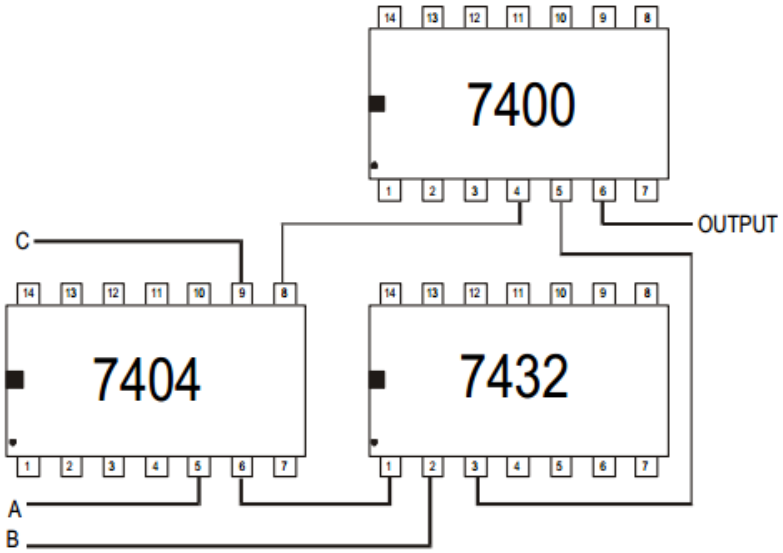


25 For the following circuit diagrams, draw the corresponding logic diagram.

a)

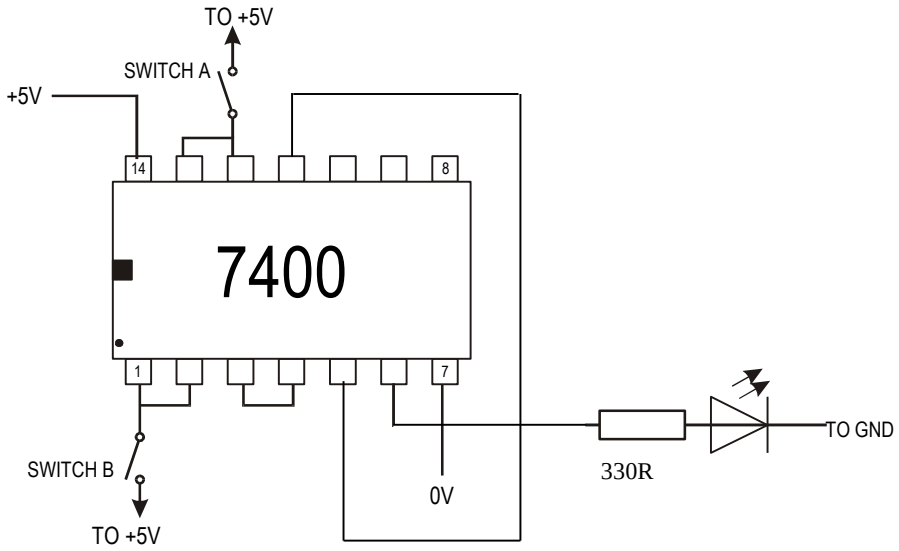


b)



26

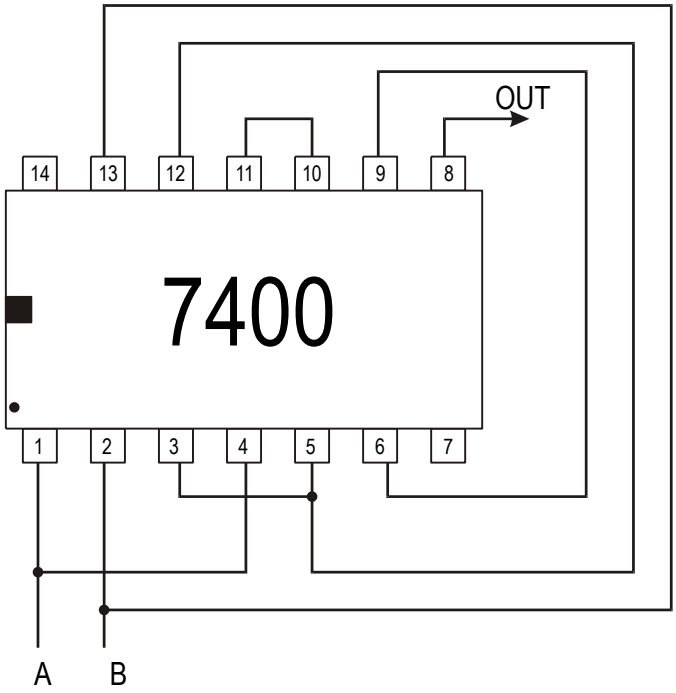
Simulate the following circuit on Yenka using two push switches as inputs and an LED to indicate the state of the output. Draw the logic gate circuit in your jotter.



Construct a truth table for this circuit and determine the single gate that could replace this NAND equivalent circuit.

27

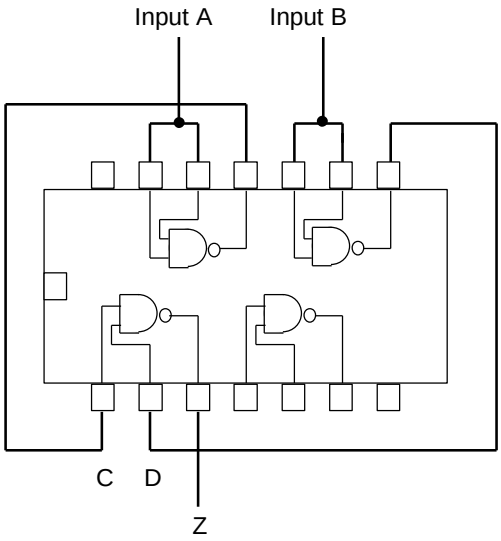
Construct the following circuit on Yenka using two push switches as inputs and a logic probe on pin 8 to indicate the state of the output. (Power supply and switch connections have been omitted for clarity.)



Construct a truth table for this circuit and determine the single gate that could replace this NAND equivalent circuit.

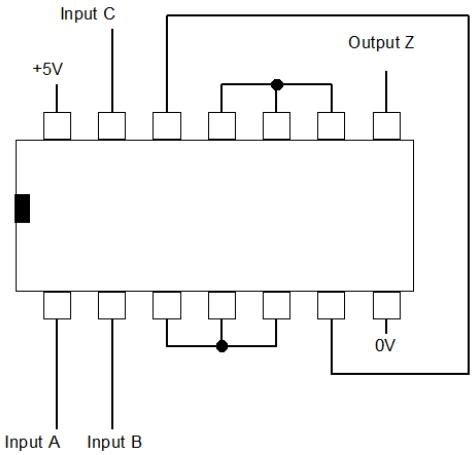
28

The diagram below shows a quadruple 2-input NAND gate IC. Construct the truth table for the circuit shown and hence state the single gate that could perform the same function. The truth table should contain columns A, B, C, D and Z.



29

a) A student constructs the circuit shown below using a 7400 IC and three input switches. Simulate the circuit on Yenka and copy the associated logic gate circuit into your jotter.



i) Draw up a truth table showing all possible input states of the three switches.

ii) Use a logic probe to monitor the output state (Z) and, using the switches, test the circuit and complete the truth table.

iii) Write a Boolean expression for this circuit and state the single gate that could replace this circuit.

b) When constructing this circuit, a student accidentally uses a 7432 IC instead of a 7400.

i) Use Yenka to construct this circuit and draw the resultant logic circuit in your jotter.

ii) Construct a truth table for this circuit and write the Boolean expression for this circuit.

iii) State the single gate that could replace this circuit and draw a simplified logic diagram using two OR gates that would perform the same function.

Exam style practice questions

1. The operation of an industrial stamping machine is controlled by a logic system with inputs and output as shown in the following table.

INPUTS	OUTPUT
L (= 1 when left-hand button is pressed)	S (= 1 to operate stamping machine)
R (= 1 when right-hand button is pressed)	
G (= 1 when guard is in position)	

The logic system shown in Figure Q1(*a*) ensures that the machine only operates when the guard is in place and either button is pressed.

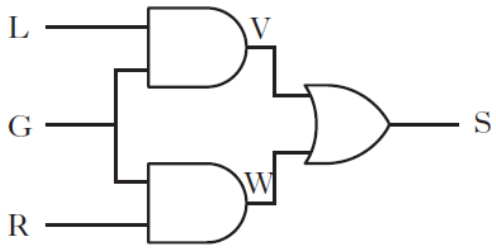


Figure Q1(*a*)

(*a*) Write a Boolean expression for the output S in terms of the inputs L, G and R. **3**

(*b*) Using only two-input NAND gates, draw a logic system equivalent to the logic system shown in Figure Q1(*a*). Delete any redundant gates. **4**

The logic system shown in Figure Q1(*b*) was designed to provide the same logic function as the logic system shown in Figure Q1(*a*).

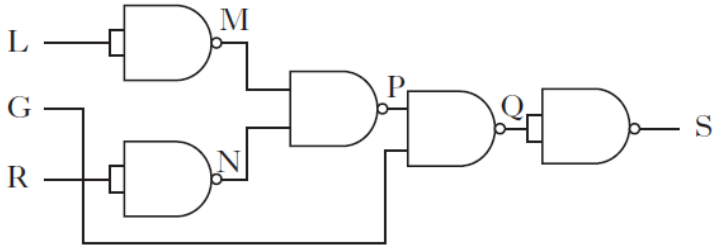


Figure Q1(*b*)

(*c*) Draw and complete a truth table for the logic system shown in Figure Q1(*b*). Include the intermediate logic values M, N, P & Q. **5**

(*d*) Draw a logic system equivalent to that shown in Figure Q1(*b*) using only two-input AND and OR gates. **3**

2. The tumble dryer shown in Figure Q1 vents warm air back into the house through a filter, and collects moisture from the clothes in a water tank. An air flow sensor detects when the filter needs to be cleaned.



Figure Q1

A logic system controls a warning buzzer. If the water tank is full (W=1) or the air flow is too low (A=0) when the start button is pressed (S=1), then the buzzer sounds (B=1).

(*a*) Draw a truth table for the output, B, in terms of the inputs S, W and A. **3**

(*b*) Write a Boolean expression for B in terms of S, W and A. **4**

(*c*) Draw a logic diagram for the control system, constructed from AND, OR and NOT gates. **3**

(*d*) Draw a logic diagram for the control system, using only NAND gates.
Simplify where appropriate. **4**

(14)